DICHOTOMY FOR LINEAR SYSTEMS IN CHARACTERISTIC P: GENERICALLY ETALE PENCIL OR SMALL-DEGREE SYSTEM ON THE FROBENIUS TWIST

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the pencil test, the all-quotients factorization with the normal-curve extension, the pullback N with its independent sections, and the full degree computation are all retained verbatim.

Editorial note. The label lemma-inseparable-linear-system names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Morphisms defined by pencils and generic etaleness. Same chapter, Lemmas lemma-linear-series and lemma-generically-etale; Proposition proposition-unwind-morphism-smooth. Identifies the rational function s/t of a pencil with the induced map to P^1, and shows that failure of generic etaleness forces f into k(X^{(p)}).
- Extending maps over normal curves. Same chapter, Lemma lemma-extend-over-normal-curve. Extends the factored map from the affine chart D_+(T_0) to all of the smooth curve X^{(p)}.
- Degree bookkeeping under relative Frobenius. Varieties, Lemmas varieties-lemma-check-invertible-sheaf-trivial, varieties-lemma-degree-pullback-map-proper-curves and varieties-lemma-inseparable-deg-p-smooth. Justifies L' = F^*N, the triviality check on components, and deg(F) = p for the relative Frobenius of a smooth curve.

Source reference. Original reference 0BXZ: sources/stacks-curves/pinned-original-source.tex, source lines 287–287. Prerequisite pointer invoked at proof line(s) 2949; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C1C: sources/stacks-curves/pinned-original-source.tex, source lines 2564–2564. Prerequisite pointer invoked at proof line(s) 2922; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CCP: sources/stacks-curves/pinned-original-source.tex, source lines 583–583. Prerequisite pointer invoked at proof line(s) 2921;

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it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CD2: sources/stacks-curves/pinned-original-source.tex, source lines 2888–2888. Prerequisite pointer invoked at proof line(s) 2934; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B40: sources/stacks-chow/tags.txt, source lines 13983–13983. Prerequisite pointer invoked at proof line(s) 2965; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYZ: sources/stacks-chow/tags.txt, source lines 13807–13807. Prerequisite pointer invoked at proof line(s) 2966; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CCF: sources/stacks-chow/tags.txt, source lines 15505–15505. Prerequisite pointer invoked at proof line(s) 2967; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field of characteristic $p > 0$. Let X be a smooth proper curve over k. Let $(\mathcal{L}, V)$ be a $\mathfrak{g}^r_d$ with $r \geq 1$. Then one of the following two is true*

1. *there exists a $\mathfrak{g}^1_d$ whose corresponding morphism $X \to \mathbf{P}^1_k$ (Lemma 0CCP) is generically étale (i.e., is as in Lemma 0C1C), or*
2. *there exists a $\mathfrak{g}^r_{d'}$ on $X^{(p)}$ where $d' \leq d/p$.*

Proof. Pick two k-linearly independent elements $s, t \in V$. Then $f = s/t$ is the rational function defining the morphism $X \to \mathbf{P}^1_k$ corresponding to the linear series $(\mathcal{L}, ks + kt)$. If this morphism is not generically étale, then $f \in k(X^{(p)})$ by Proposition 0CD2. Now choose a basis $s_0, \ldots, s_r$ of V and let $\mathcal{L}' \subset \mathcal{L}$ be the invertible sheaf generated by $s_0, \ldots, s_r$. Set $f_i = s_i/s_0$ in $k(X)$. If for each pair (s_0, s_i) we have $f_i \in k(X^{(p)})$, then the morphism

$$\varphi = \varphi_{(\mathcal{L}', (s_0, \ldots, s_r))} : X \longrightarrow \mathbf{P}^r_k = \mathrm{Proj}(k[T_0, \ldots, T_r])$$

factors through $X^{(p)}$ as this is true over the affine open $D_+(T_0)$ and we can extend the morphism over the affine part to the whole of the smooth curve $X^{(p)}$ by Lemma 0BXZ. Introducing notation, say we have the factorization

$$X \xrightarrow{F_{X/k}} X^{(p)} \xrightarrow{\psi} \mathbf{P}^r_k$$

of φ. Then $\mathcal{N} = \psi^*\mathcal{O}_{\mathbf{P}^1_k}(1)$ is an invertible $\mathcal{O}_{X^{(p)}}$-module with $\mathcal{L}' = F^*_{X/k}\mathcal{N}$ and with $\psi^*T_0, \ldots, \psi^*T_r$ k-linearly independent (as they pullback to $s_0, \ldots, s_r$ on X). Finally, we have

$$d = \deg(\mathcal{L}) \geq \deg(\mathcal{L}') = \deg(F_{X/k}) \deg(\mathcal{N}) = p \deg(\mathcal{N})$$

as desired. Here we used Varieties, Lemmas 0B40, 0AYZ, and 0CCF. □